```python
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score
iris = load_iris()
X = iris.data
y = iris.target
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=1
)
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
model = MLPClassifier(
    hidden_layer_sizes=(5,),
    max_iter=1000,
    random_state=1
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
print("Artificial Neural Network using Backpropagation")
print("-----------------------------------------------")
print("Training Samples :", len(X_train))
print("Testing Samples  :", len(X_test))
print("Accuracy         :", round(accuracy_score(y_test, y_pred)*100,2),
"%")
print("\nFirst 10 Predictions")
print("Predicted :", y_pred[:10])
print("Actual    :", y_test[:10])
new_sample = [[5.1, 3.5, 1.4, 0.2]]
new_sample = scaler.transform(new_sample)
prediction = model.predict(new_sample)
```

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Artificial Neural Network using Backpropagation
-----------------------------------------------
Training Samples : 105
Testing Samples  : 45
Accuracy         : 95.56 %

First 10 Predictions
Predicted : [0 1 1 0 2 1 2 0 0 2]
Actual    : [0 1 1 0 2 1 2 0 0 2]

New Sample Prediction:
Predicted Flower : setosa
/usr/local/lib/python3.12/dist-packages/sklearn/neural_network/_multilayer_perceptron.py:
  warnings.warn(
```

```
print("\nNew Sample Prediction:")
print("Predicted Flower :", iris.target_names[prediction[0]])
```